

UNIT – 1
ASSESSMENT – 2
PROBLEMS

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Question 1

Scenario

A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is dynamic, with blocked roads, changing weather, and varying travel costs. The drone uses heuristic estimates and real-time updates to find the safest and fastest route.

Tasks

1. Explain the behavior of Greedy Best-First Search and its strengths and weaknesses.
2. Explain how A* balances cost and heuristic.
3. Analyze the impact of heuristic accuracy.
4. Discuss how blocked paths affect search performance.
5. Recommend the most suitable algorithm with justification.

Question 2

Scenario

A smart city uses an AI system to optimize traffic signal timings across hundreds of intersections. The goal is to minimize waiting time, fuel consumption, and traffic congestion. The system must continuously adjust signal timings based on changing traffic conditions. Since the search space is very large and traffic patterns change dynamically, the system may get stuck in locally optimal solutions.

Tasks

1. Formulate the problem as an optimization problem and explain why traditional search is inefficient.
2. Explain how Hill Climbing works and identify its limitations.
3. Discuss local maxima, plateaus, and ridges with real-world examples.
4. Explain how Simulated Annealing or Genetic Algorithms overcome these limitations.
5. Recommend the best approach with justification.

Question 3

Scenario

An autonomous rover is deployed on Mars to explore unknown terrain. It must navigate safely to collect samples, avoid hazards such as craters and rocks, operate with limited sensing capability, and make decisions

without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks

1. Explain why this problem requires an online search agent instead of offline search.
2. Analyze the challenges of partial observability and dynamic environments.
3. Describe how the rover builds and updates its internal model.
4. Compare online search with uninformed and informed search strategies.
5. Justify the most suitable approach considering uncertainty, adaptability, and safety.

Question 4

Scenario

A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints such as:

- No student should have overlapping exams.
- Limited number of exam halls.
- Certain subjects must be scheduled before others.
- Faculty availability constraints.

- Special accommodations for some students.

As the number of students and courses increases, the scheduling problem becomes more complex.

Tasks

1. Formulate the problem as a Constraint Satisfaction Problem (CSP).
2. Define variables, domains, and constraints.
3. Explain how Backtracking Search works.
4. Explain how Forward Checking improves efficiency.
5. Analyze real-world challenges and recommend the best strategy.

Question 5

Scenario

A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, make decisions within strict time limits, evaluate complex game states with many possible moves, and handle a very large decision tree.

Tasks

1. Explain how the Minimax algorithm models this problem.
2. Analyze the computational challenges of large game trees.

3. Explain how Alpha-Beta Pruning improves efficiency.
4. Design an evaluation function and justify the factors considered.
5. Recommend the best strategy for balancing optimality and real-time performance.